
Monash University, Faculty of Information Technology
FIT3080**Lab : Reinforcement Learning****Sample Solutions****Exercise 1: Multi-Armed Bandit**

Consider a multi-armed bandit problem with $k = 4$ actions, denoted 1, 2, 3, and 4. Consider applying to this problem a bandit algorithm using ϵ -greedy action selection, sample-average action-value estimates,

$$Q_n(a_k) = \frac{R_1(a_k) + R_2(a_k) + \dots + R_{n-1}(a_k)}{n(a_k)},$$

and initial estimates of $Q(a_k) = 0$ for all a . Note n represents time/iteration index and $n(a_k)$ is the number of times this action receives rewards. Suppose the initial sequence of actions and rewards is:

at time 1 $a_k = 1$ $R_1(1) = 1$

at time 2 $a_k = 2$ $R_2(2) = 1$

at time 3 $a_k = 2$ $R_3(2) = 2$

at time 4 $a_k = 2$ $R_4(2) = 2$

at time 5 $a_k = 3$ $R_5(3) = 0$

On some of these time steps the ϵ case may have occurred causing an action to be selected at random. On which time steps did this occur? On which time steps could this possibly have occurred?

Possible Solution: The sequence of values for tuples S_1, S_2, S_3, S_4 , at each time-step is presented below:

- Time 1: 0, 0, 0, 0
- Time 2: 1, 0, 0, 0
- Time 3: 1, 1, 0, 0
- Time 4: 1, 1.5, 0, 0
- Time 5: 1, 1.67, 0, 0

The sequence of steps associated to the certainty of ϵ having occurred is shown next:

Time 1: Maybe: $Q(a_k) = 0 \forall k$ at this point, so choosing $A_1 = 1$ could have happened as a random choice due to ϵ , or by randomly choosing an action when tie-breaking.

Time 2: Yes: $Q(a_1) > 0$ at this point, so choosing $A_2 = 2$ is necessarily a random decision, otherwise A_1 would have been chosen.

Time 3: Maybe: $Q(a_k) = 1$ for $k \in \{1, 2\}$ at this point, so choosing $A_3 = 2$ could have happened due to ϵ , or by randomly choosing an action when tie-breaking.

Time 4: Maybe: $Q(a_2) = 1.5$ at this point, so choosing $A_4 = 2$ could have happened due

to ϵ , or by selecting the option action with highest Q at this point, which is a_2 .

Time 5: Yes: $Q(a_2) = 1.67$ is the highest Q-value at this point, so choosing $A_5 = 3$ is necessarily a random decision, otherwise A_2 would have been chosen.

Exercise 2: Q Learning

Consider a system with two states and two actions. You perform actions and observe the rewards and transitions listed below. Each step lists the current state, reward, action and resulting transition as $S_i; R = r; a_k : S_i \rightarrow S_j$. Perform Q-learning using a learning rate of $\alpha = 0.5$ and a discount factor of $\gamma = 0.5$ for each step. The Q-table entries are initialized to zero.

$$Q(a, s) \leftarrow Q(a, s) + \alpha(R(s) + \gamma \max_{a'} [Q(a', s')] - Q(a, s))$$

Possible Solution:

1) S_1 $R = -10$ $a_1 : S_1 \rightarrow S_1$

Q	S_1	S_2
a_1	-5	0
a_2	0	0

$$\begin{aligned} Q(a_1, S_1) &\leftarrow 0 + 0.5(-10 + 0.5 \max_{a'} Q(a', S_1) - 0) \\ &= 0 + 0.5(-10 + 0.5 \times 0) \\ &= -5 \end{aligned}$$

2) S_1 $R = -10$ $a_2 : S_1 \rightarrow S_2$

Q	S_1	S_2
a_1	-5	0
a_2	-5	0

$$\begin{aligned} Q(a_2, S_1) &\leftarrow 0 + 0.5(-10 + 0.5 \max_{a'} Q(a', S_2) - 0) \\ &= 0 + 0.5(-10 + 0.5 \times 0) \\ &= -5 \end{aligned}$$

3) S_2 $R = +20$ $a_1 : S_2 \rightarrow S_1$

Q	S_1	S_2
a_1	-5	8.75
a_2	-5	0

$$\begin{aligned} Q(a_1, S_2) &\leftarrow 0 + 0.5(20 + 0.5 \max_{a'} Q(a', S_1) - 0) \\ &= 0 + 0.5(20 + 0.5 \times (-5)) \\ &= 8.75 \end{aligned}$$

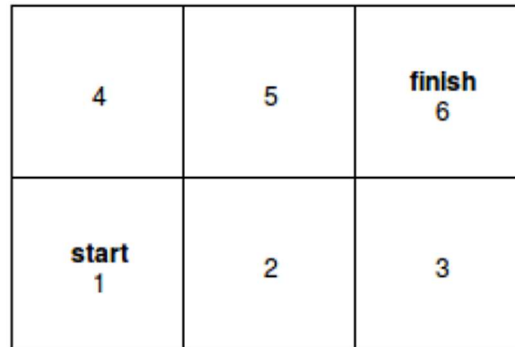
4) S_1 $R = -10$ $a_2 : S_1 \rightarrow S_2$

Q	S_1	S_2
a_1	-5	8.75
a_2	-5.3125	0

$$\begin{aligned} Q(a_2, S_1) &\leftarrow -5 + 0.5(-10 + 0.5 \max_{a'} Q(a', S_2) - (-5)) \\ &= -5 + 0.5(-10 + 0.5 \times 8.75 + 5) \\ &= -5 + 0.5(-0.625) \\ &= -5.3125 \end{aligned}$$

Exercise 3: SARSA

Assume we are an agent in a 3x2 grid world, as shown in the below figure. We start at the bottom left node (1) and finish in the top right node (6). When node 6 is reached, we receive a reward of +10 and return to the start for a new episode. On all other actions that will not lead to state 6, the reward is -1.



In each state we have four possible actions: **up, down, left and right**. For each action we move deterministically in the specific direction on the grid. Assume that we cannot take actions that bring us outside the grid.

The current estimates of Q (s, a) are given in the below table:

Q(1,up)=4			Q(1,right)=3
Q(2,up)=6		Q(2,left)=3	Q(2,right)=8
Q(3,up)=9		Q(3,left)=7	
	Q(4,down)=2		Q(4,right)=5
	Q(5,down)=6	Q(5,left)=5	Q(5,right)=8

We will continue updating the Q-table with a SARSA (state-action-reward-state-action) algorithm. Starting from node 2, we have sampled the following trajectory: **2 - up - 5 - right - 6**, after which the trial ended. Update two Q(s, a) entries by filling in the form below (take $\alpha = 0.2$; $\gamma = 0.8$). The SARSA equation is provided:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha[r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)]$$

Hint: The action-value of a terminal state is 0.

Solution:

s	a	r	s'	a'
2	up	-1	5	right

$$Q(2, \text{up}) = 6 + 0.2 * [-1 + 0.8 * (8) - 6] = 6 - 0.12 = 5.88$$

s	a	r	s'	a'
5	right	10	6	- (episode terminated)

$$Q(5, \text{right}) = 8 + 0.2 * [10 + 0.8 * (0) - 8] = 8 + 0.4 = 8.4$$

Exercise 4: *Q-Learning for the Maze (Python Code)*

In this exercise, you are required to implement the *Q-Learning* algorithm for the same maze as the last lab. The maze has two terminal states (green and red blocks), but there are small negative rewards, $R = -0.1$, on the non-terminal states. You need to write a Python code that performs the following tasks:

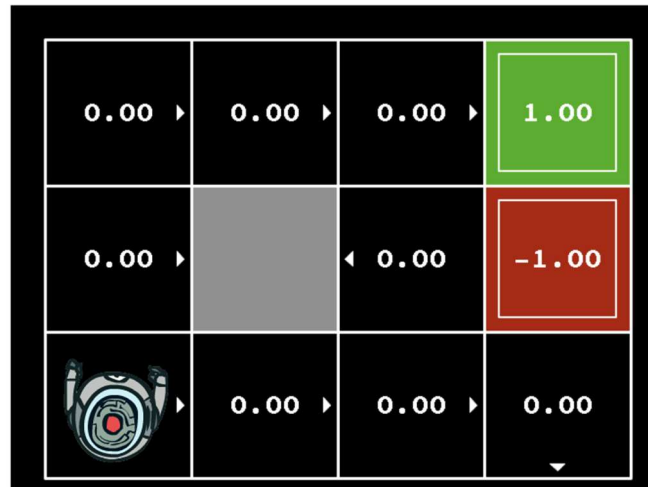


Figure 1: Maze Structure

1. Implement the Q-Learning algorithm with a learning rate α of 0.6 and a discount factor γ of 0.9.
2. Set the start position for the agent at the bottom-most left block of the maze (suggested coordinate: (3, 0)).
3. Initialise the Q-table entries by zero.
4. Calculate the Q-values and policy for each block of the maze.
5. Try different initial blocks of your choice and record your observations.

Solution: see [associated code](#).